

S T R O M E X

E X E C U T I V E K N O W L E D G E S Y S T E M

B O O K I V

IV

Engineering, AI Architecture, Cloud & Security

How it is built · How it stays up · How it stays safe

The StromeX Editorial Bible · Edition II

AUTHORITY — OPERATIONAL — CTO

13 AUGUST 2026 · VERSION 2.0

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EXECUTIVE SUMMARY

Volume IV governs construction. It sets ten engineering principles, the canonical stack, and the static-site pattern that four reference implementations converged on independently. It specifies multi-tenancy with the no-fork rule, the data architecture and its classification scheme, and — in the most consequential subsystem the group operates — the credential and verification architecture, including the signing model in which StromeX never signs on an institution's behalf. It defines the AI architecture and its provider-independence doctrine, the stakes ladder that determines when a human must decide, the data rights and model-training policy, the performance budgets and degraded modes required by constrained environments, and internationalisation engineering. Chapters twelve to seventeen constitute the Security Bible, covering zero trust, identity, cryptography, secure development, resilience and incident response.

One book of ten. The corpus index and the other nine are at [docs/library/](#).

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VOLUME

IV

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CHAPTER 1 — ENGINEERING PRINCIPLES

E1 — Boring beneath, ambitious above. Payments, authentication, credential signing, backups and audit logs use conservative, proven, widely-reviewed technology. Novelty budget is spent on the customer-visible frontier. A clever database is a liability; a clever admissions experience is an asset.

E2 — Build for the constrained case first. A 5-year-old Android on 3 Mbps with intermittent power is the reference environment. If it works there, it works everywhere. The inverse is never true.

E3 — One platform, many surfaces. Layers 1–4 of the ecosystem stack (Volume I §2.3) are shared by every sector. Forking them is a governed exception (Chapter 3), not an engineering convenience.

E4 — Reversibility over velocity. Every deployment rolls back. Every migration reverses. Every destructive user action has a recovery window. We ship faster *because* mistakes are cheap, not despite it.

E5 — Data is the customer's. Export is complete, open-format, self-service, free and immediate. This is enforced in code, in tests, and in the contract.

E6 — Correctness beats cleverness in the record. The system of record is the moat (Volume II §9). An elegant abstraction that risks a wrong grade, a wrong fee balance, or a wrongly-issued certificate is not elegant.

E7 — Everything is auditable. Who did what, to which record, when, from where, and what it was before. Not a feature — a property of the platform.

E8 — Observable before scalable. We do not optimise what we cannot measure, and we do not ship what we cannot see failing.

E9 — Composition over configuration over customisation. Prefer combining existing modules; then configuring them; then, last and reluctantly, writing bespoke code for one customer.

E10 — The build must be reproducible by a new hire on day two. If setup requires tribal knowledge, that is a defect with a ticket.

CHAPTER 2 — THE REFERENCE ARCHITECTURE

2.1 The shape

SURFACES	Web · PWA · Android · iOS · Kiosk · Signage · USSD · Print · Card · Gate · API consumers
EDGE	CDN · WAF · DDoS · rate limit · geo-routing
BFF	Per-surface aggregation, response shaping, caching
DOMAIN SERVICES	Identity · Records · Academic · Finance · Credentials · Comms · Automation · Intelligence
PLATFORM	AuthN/Z · Audit · Search · Files · Events · Jobs · Notifications · Payments · Tenancy · Feature flags
AI	Router · Agents · RAG · Guardrails · Evaluation
DATA	PostgreSQL · Object store · Vector · Search · Cache · Queue · Warehouse
INFRA	Compute · Network · Secrets · Backup · Observ.

Figure 1 — The shape

2.2 The canonical stack

Chosen for hiring depth in our markets, operational simplicity, and longevity — not for novelty.

LAYER	CHOICE	WHY
Primary database	PostgreSQL	Relational integrity for a record system; JSONB where flexibility is genuinely needed; enormous operational knowledge base
API services	Python (FastAPI) for domain services	Already the MVP's stack; strong AI/ML adjacency; fast to hire for
Web	Next.js / TypeScript / Tailwind	Already the MVP's stack; SSR for performance on weak devices; huge talent pool
Static institutional sites	Token-driven static build	The pattern proven across four reference sites: partials + a page manifest + a build script; no framework, no runtime, near-zero hosting cost, trivially cacheable, survives a decade
Mobile	Capacitor wrapper first, native where justified	Already delivered; one codebase; native only where the platform genuinely requires it
Vector store	Qdrant	Already in the MVP

LAYER	CHOICE	WHY
Cache / queue	Redis	Already in the MVP
Search	PostgreSQL FTS , then a dedicated cluster at scale	Do not add an engine before the query volume justifies it
Containers	Docker / Compose → orchestrated at scale	Already in infra/
Edge	Cloudflare	Already the production topology

Table 1 — The canonical stack

The two-stack rule: the group runs at most two primary backend languages and two frontend frameworks at any time. A third requires CTO approval and a written retirement plan for one of the existing two.

2.3 The static-site pattern, promoted to a product

The four institutional reference sites converged independently on the same architecture, which is now the group standard for institutional web presence:

- [partials/](#) — shared chrome as language-neutral templates carrying `{{t:key}}` translation tokens; **one file serves every language**
- [pages/](#) — page body content only, one file per page per locale
- [pages/manifest.json](#) — the registry: slug, output path, title, description, content file, `lang`, `dir`, `altHref`
- [i18n/<code>.json](#) — one flat, dot-keyed dictionary per language
- [i18n/locales.json](#) — the single source of truth for which languages exist
- [scripts/build.js](#) — assembles partials + content into complete documents, deriving untranslated locales with translated chrome and a labelled notice rather than emitting a broken URL

Why this is the standard. It produces a four-language site whose marginal cost per additional language is a dictionary rather than a codebase; it has no runtime and therefore no runtime vulnerabilities; it is trivially CDN-cacheable, which is decisive on 3 Mbps connections; it costs almost nothing to host, which is what makes the Foundation tier affordable; and it will still build in ten years because it depends only on the language’s standard library. This is the architectural expression of principle E1.

Never do: hand-edit built output. Regenerate.

CHAPTER 3 — MULTI-TENANCY & THE NO-FORK RULE

3.1 The tenancy model

Shared schema with a mandatory tenant discriminator, enforced at the data layer rather than trusted to application code, with row-level security as the backstop. Dedicated schemas or dedicated databases are available at Elite/Enterprise/Government tiers where residency, regulation or scale require it.

MODEL	WHO GETS IT	COST TO US
Shared schema, RLS-enforced	Foundation → Premium	Lowest
Dedicated schema	Elite, large multi-campus	Moderate
Dedicated database	Enterprise, regulated	High
Dedicated deployment	Government, air-gapped	Highest (Volume III §4.6)

Table 2 — The tenancy model

Non-negotiable: a tenant-scoped query that could return another tenant's row is a **P0 security incident**, not a bug. Every data-access path has a test that attempts cross-tenant access and asserts failure. This test suite may never be skipped in CI.

3.2 The no-fork rule

Forking platform layers 1–4 for a single customer requires written CTO approval, a documented justification, and a **sunset date**. Without a sunset date the answer is no.

The reason is arithmetic: breadth across twenty sectors is affordable only because the lower layers are shared (Volume I §2.3). Every fork multiplies the maintenance surface by the number of forks, and forks never get deleted unless someone was made responsible for deleting them on a date.

Legitimate alternatives, in order of preference: a feature flag, a configuration option, a tenant-scoped extension point, a plugin, a sector module, and only then a fork.

CHAPTER 4 — DATA ARCHITECTURE

4.1 The record principles

D1 — One institution, one person record. A student who becomes an alumnus who becomes a parent who becomes a staff member is one person with four relationships, not four records. This is harder and it is correct.

D2 — Events, not just state. Enrolment, role changes, fee movements, credential issuance and governance decisions are recorded as **events with a time**, from which state is derived. Auditability, reversibility and historical reconstruction all fall out of this for free, and none of them can be retrofitted.

D3 — Soft delete by default, hard delete on request. Nothing is destroyed accidentally; everything is destroyable deliberately, verifiably, and in fulfilment of a data subject's rights.

D4 — Every record carries provenance. Created by, modified by, source system, and confidence where derived.

D5 — Reference data is versioned. Curricula, fee structures, grading scales and policies change between years. A 2029 transcript must render under 2029's grading scale, not today's.

4.2 The canonical domain model

Validated against the 73-table credential-and-commerce schema in the group's reference implementation, which covers: users and roles; academic bodies, programmes, levels, courses, units and learning outcomes; enrolments and enrolment-integrity events; competencies, skills, assessments and marks; evidence items and versions; quizzes, exercises, vocabulary and self-checks; recordings, pronunciation targets and feedback; time-on-task and unit progress; awards, distinctions and award verifications; credential signatures, signing keys and issued documents; verifying institutions and institution checks; graduate profiles, profile shares and alumni chapters; payments, receipts, refunds, instalment plans, promo codes, scholarships, currencies and country payment routing; corporate accounts and seats; CPD records; live sessions; notifications.

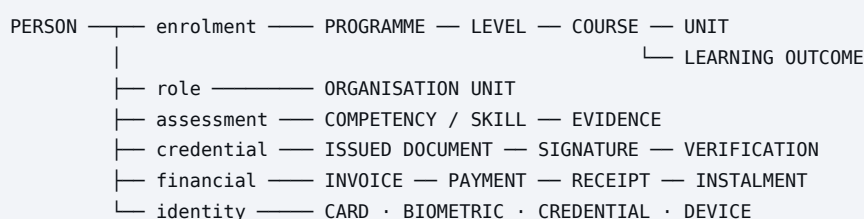


Figure 2 — The canonical domain model

4.3 Data classification

CLASS	EXAMPLES	HANDLING
Public	Published pages, verification results	CDN-cacheable
Internal	Aggregate statistics, non-identifying config	Authenticated
Confidential	Person records, grades, finance	Encrypted at rest, access-logged, least-privilege

CLASS	EXAMPLES	HANDLING
Sensitive	Health, safeguarding, discipline, biometrics	Additionally: field-level encryption, named-role access only, every read logged and reviewable
Children’s data	Any record of a person under 18	Strictest class in the group. Minimisation, short retention, no behavioural profiling, no advertising use, parental access rights, and a standing prohibition on any secondary use

Table 3 — Data classification

4.4 Retention

Defaults, overridable upward by regulation and downward by customer instruction: operational logs 90 days · audit logs 7 years · financial records per jurisdiction (typically 6–7 years) · academic records **permanent by default** (a transcript must be producible in forty years) · biometric templates deleted on relationship end · marketing data 24 months from last engagement · AI conversation logs 12 months, or 30 days where they contain sensitive-class data.

CHAPTER 5 — THE CREDENTIAL & VERIFICATION ARCHITECTURE

The single most strategically important subsystem in the group (Volume II §9). Specified in detail because getting it wrong is unrecoverable — a credential system that is found to be forgeable destroys the institution’s trust and ours simultaneously.

5.1 The four requirements

1. **Unforgeable** — a credential cannot be fabricated by anyone without the issuer’s private key.
2. **Verifiable by anyone, free, forever** — the constitutional commitment (Volume I §5.3).
3. **Verifiable offline where possible** — a border officer with no connectivity should still get signal.
4. **Revocable** — issuance is not irreversible; a credential obtained fraudulently must be withdrawable and the withdrawal must be visible.

5.2 The identifier architecture

Every credential carries a **structured, human-transcribable identifier** designed with the institution (a service line — Volume III §3.1). It encodes issuing institution, document class, year, and sequence, with a check character. It must be readable aloud over a phone call without ambiguity, which rules out characters that collide when spoken or handwritten (0/O, 1/I/l, 5/S, 8/B).

5.3 The signing model

- Each issuing institution holds a distinct key pair. **StromeX never signs on an institution’s behalf** — the institution is the issuer, in fact and in cryptography.
- Private keys live in a hardware security module or an equivalent managed key service. Key material is never in application memory longer than a signing operation, never in logs, never in backups in plaintext.
- Key generation, custody, rotation and revocation follow a documented ceremony with two-person control and a written record. This is a priced service (Volume III §3.1) because it is real work with real consequences.
- Compromise of one institution’s key affects that institution only. There is no group-wide master key, deliberately — a single key whose compromise invalidates every credential the group ever issued is an unacceptable concentration of risk.

5.4 The verification surfaces

Four distinct surfaces are in production across the reference implementations — certificate, identity, receipt and graduation document — because they have genuinely different verification semantics and conflating them produces confusing results.

SURFACE	ANSWERS
Certificate verification	Was this award made, to this person, by this institution, on this date, and is it still valid?
Identity verification	Is this person currently who this card says they are, in this role?
Receipt verification	Was this payment actually received and applied?
Document verification	Is this multi-page document the sealed original, unaltered?

Table 4 — The verification surfaces

Every verification response states plainly: **valid**, **revoked** (with date and reason class), **not found**, or **superseded** — never an ambiguous result, and never a bare “invalid” that fails to distinguish a forgery from a typo.

5.5 The layered verification model

LAYER	MECHANISM	WORKS OFFLINE	STRENGTH
1	Human-readable identifier	Yes	Weak alone
2	QR → verification URL	No	Strong, convenient
3	Detached digital signature embedded in the document	Yes	Strong
4	Physical security features (guilloche, microtext, UV, foil)	Yes	Deters casual forgery
5	Optional public-ledger anchor	Yes	Independent of us existing

Table 5 — The layered verification model

Layer 5 exists for one reason and it is an honest one: it lets an institution’s credentials remain verifiable even if StromeX ceases to exist. Offering it costs us a theoretical lock-in advantage and buys the customer real assurance. That trade is required by Volume I P5.

5.6 Revocation

A public, append-only revocation register. Revocation reasons are classified (issued in error, obtained fraudulently, superseded, withdrawn by institution) and the class is public while the narrative is not. Revocation requires a documented approval chain — one person must not be able to revoke a graduate’s degree alone.

CHAPTER 6 — API & INTEGRATION STANDARDS

STANDARD	RULE
Style	REST + JSON. GraphQL only where a surface genuinely needs client-shaped queries
Versioning	URL-versioned (<code>/v1/</code>). A version is supported for 24 months after its successor ships
Breaking changes	Never within a version. New version, dual-run, deprecation notice, telemetry-verified migration
Auth	OAuth 2.1 / OIDC; API keys for server-to-server; short-lived tokens; refresh rotation
Pagination	Cursor-based. Offset pagination is prohibited on any table that can exceed 100k rows
Errors	Structured: machine code, human message, remediation hint, correlation ID
Rate limits	Published, returned in headers, generous by default
Idempotency	Mandatory on every state-changing endpoint. Payments without idempotency keys are rejected
Webhooks	Signed, retried with exponential backoff, replayable for 7 days, with a delivery log the customer can see
Documentation	OpenAPI, generated from code, published, with runnable examples. Free and public
SDKs	JS/TS, Python, PHP. Open source. Generated where possible, hand-polished where not
Deprecation	Announced, dated, telemetry-monitored, and never enforced before the announced date

Table 6 — Api & Integration Standards

The dogfooding rule: every StromeX surface consumes the same public API a third party would. There is no privileged internal API. If it is awkward for us, it is awkward for them, and we will find out first.

CHAPTER 7 — THE AI ARCHITECTURE

7.1 The provider-independence doctrine

No StromeX capability may depend on a single AI provider. This is a commercial risk control (Volume II R7) as much as an architectural preference: provider pricing, availability, terms and model behaviour all change without our consent.

The MVP already implements multi-provider routing across Claude, OpenAI-compatible endpoints, DeepSeek and Perplexity, with a development provider for offline work. That router is the group standard and every AI capability goes through it.

```
Request → Policy gate → Router → Provider (primary)
                                |
                                └─ fallback chain
                                └─ Cost / latency / capability routing
                                └─ Self-hosted model (for sensitive classes)
                                └─ Deterministic fallback (never a blank failure)
```

Figure 3 — The provider-independence doctrine

Routing criteria, in order: data classification (sensitive-class work may be pinned to a self-hosted or residency-compliant model) → required capability → latency budget → cost. Cost is last, always. A cheaper model that gets a grade wrong is not cheaper.

7.2 Retrieval before generation

Institution-facing answers are **grounded in the institution’s own records and documents**, retrieved and cited, not recalled from a model’s training. The pattern:

```
Query → tenant-scoped retrieval → relevance filter → context assembly
      → generation with mandatory citation → citation validation
      → confidence assessment → response, or escalation
```

Figure 4 — Retrieval before generation

Citation validation is a real check, not a prompt instruction. A generated citation that does not resolve to a retrieved source is stripped, and if stripping leaves the answer unsupported the system says it does not know. Volume I §7.2.3 prohibits fabricated authority, and prompts alone do not enforce prohibitions.

7.3 The agent model

An agent is a **bounded role**, not a personality: a scope of data it may read, a set of actions it may take, a refusal boundary, an escalation target, and an audit trail. Agents are configured per institution, and the institution can see and change every one of those five things.

PROPERTY	RULE
Data scope	Explicit allow-list. An agent cannot read what its role would not read
Actions	Explicit allow-list. Write actions require a configured approval where they affect a person’s record
Refusal boundary	Per-domain, enforced in the policy gate, not in the prompt

PROPERTY	RULE
Escalation	Every agent has a named human target. An agent with no escalation path may not be deployed
Audit	Every invocation logged: prompt, retrieved context IDs, model, output, action taken, cost
Kill switch	Per-agent, per-institution, immediate, available to the institution

Table 7 — The agent model

7.4 Evaluation

Agents are evaluated before release and continuously in production against: **factual accuracy** (against a held-out set of the institution's real records), **citation validity**, **refusal correctness** (does it decline what it must decline), **consistency** (repeated identical queries must not vary factually), **latency**, and **cost per task**.

A model or prompt change ships only when the evaluation suite is at or above the incumbent on accuracy, citation validity and refusal correctness. Latency and cost improvements do not license accuracy regressions.

CHAPTER 8 — AI GOVERNANCE & HUMAN-IN-THE-LOOP ENFORCEMENT

8.1 The stakes ladder

STAKES	EXAMPLES	RULE
Low	Draft an email, summarise a document, suggest a lesson activity	AI acts; human may review
Medium	Mark homework, draft a policy, propose a timetable, triage an enquiry	AI proposes; human approves before it reaches the affected person
High	Final grade, admission decision, disciplinary outcome, fee write-off, safeguarding classification, staff performance outcome	AI may inform; a named human decides and is recorded as the decider
Prohibited	Religious rulings, legal conclusions, medical diagnosis, immigration advice, credential issuance without human authorisation	AI does not decide. It routes to a qualified human and says so to the user

Table 8 — The stakes ladder

This ladder is enforced in the policy gate. A capability that would breach it cannot be configured into existence, and an institution cannot opt out of the High and Prohibited rows — not even by request, and not for a discount. Volume I P10 and §7.2.4.

8.2 Disclosure

Every AI surface is labelled as AI, in the user’s language, at the point of interaction — not in a footer, not in a terms page. Where a deliverable is produced with AI assistance and human review, both facts are stated. Where a human reviewed it, that human is identifiable internally and accountable for it.

8.3 Faith and cultural content

Qur’anic text is **retrieved from verified sources and reproduced verbatim** — never generated, never paraphrased, never completed by a model. Translations and tafsir are attributed to their named scholars and editions. Questions of ruling route to qualified human scholars with the routing visible to the user. The same standard of fidelity applies to every tradition the group serves; there is no primary tradition and no set of afterthoughts (Volume I §7.5).

8.4 The prohibited-use enforcement

Prompt injection, jailbreak attempts and scope-escape attempts are detected, logged, and rate-limited. Repeated attempts against a sensitive-class agent trigger a human review of the account. Users are told when a request was declined and why, in plain language, without a lecture.

CHAPTER 9 — DATA RIGHTS & MODEL TRAINING POLICY

Stated as flatly as possible because ambiguity here is worth more to a competitor than any feature:

5. **Customer data is the customer's.** StromeX is a processor.
6. **We do not train third-party models on customer data.** Not with de-identification, not with aggregation, not with consent buried in terms. Where a provider's default terms would permit training, we contract out of it, and where we cannot, we do not route customer data to that provider.
7. **We do not sell, broker, licence or share customer data.** No exceptions for “partners”, “analytics”, or “research”.
8. **Aggregate benchmarking is opt-in, anonymised, and never leaves the group** (Volume III Division 9).
9. **We may train our own models on our own operational data** — our documentation, our code, our support transcripts where the customer's content is removed — and on data a customer has explicitly and separately licensed to us, with the licence revocable and the benefit shared.
10. **Sub-processors are published**, with the data classes they receive, and customers are notified before a new one is added.
11. **Export is complete, open-format, self-service and free**, at any time, including after termination, for a defined post-termination window that is stated in the contract and is never shorter than 90 days.

CHAPTER 10 — PERFORMANCE & THE CONSTRAINED-ENVIRONMENT STANDARD

10.1 Budgets — release blockers, not aspirations

METRIC	TARGET	MEASURED ON
First contentful paint	< 1.5s	3 Mbps, mid-range Android
Largest contentful paint	< 2.5s	same
Time to interactive	< 3.5s	same
Interaction to next paint	< 200ms	same
Cumulative layout shift	< 0.1	same
API p50 / p95 / p99	120ms / 400ms / 900ms	production
Initial JS payload	< 180 KB gzipped	per route
Total page weight	< 900 KB	first load, incl. images
Fonts	≤ 4 files, subset, <code>font-display: swap</code>	—

Table 9 — Budgets — release blockers, not aspirations

10.2 The degraded-mode requirement

Every surface has a **defined behaviour under failure**, designed and tested, not emergent:

FAILURE	REQUIRED BEHAVIOUR
No network	Cached read access; queued writes; explicit “not synced” state; nothing silently lost
Slow network	Progressive rendering; skeleton states; no blocking spinners over 3s
No power (device)	Work is persisted continuously; nothing depends on a clean shutdown
Payment provider down	Queue and retry; offer alternative channel; never take money without recording it
AI provider down	Fall back through the chain; then a deterministic non-AI path; never a blank error
Our platform down	Static institutional site stays up (it has no runtime); status page states truth

Table 10 — The degraded-mode requirement

Offline-first is not a feature flag. Service workers, local persistence and sync-conflict resolution are architectural, and the reference implementations already ship them.

10.3 Data-cost awareness

In our founding markets users pay for data by the megabyte. A heavy page is a real cost transferred to a parent. Image budgets, lazy loading, and a “low-data mode” that ships text-first are therefore engineering requirements, not optimisations.

CHAPTER 11 — FRONTEND, DESIGN SYSTEM & INTERNATIONALISATION ENGINEERING

11.1 The design system

StromeX Design System — open source (Volume I §5.5) — is the single source of truth: tokens (colour, type, space, radius, motion, elevation), primitives, components, patterns, and per-sector theme layers. Divisions extend it; they do not fork it. A component used by three products is promoted into the system, and the third team to need it does the promotion.

Accessibility is enforced mechanically: contrast ratios checked in CI, keyboard traversal tested, focus visible always, **prefers-reduced-motion** honoured, semantic HTML before ARIA, and every interactive element reachable and labelled. WCAG 2.2 AA is the floor and it is verified, not asserted.

11.2 Internationalisation engineering

The group ships English, Arabic, Yorùbá and French today across its reference implementations. The standards that made that affordable:

12. **One dictionary per language**, flat and dot-keyed. Nested structures produce merge conflicts and orphaned keys.
13. **A locale registry as the single source of truth** for which languages exist and their direction.
14. **Language-neutral chrome templates** carrying translation tokens — one file per component, not one per language. Per-language copies are technical debt with a migration path, and the reference implementations record exactly this transition.
15. **Bidirectional layout is a property, not a mode.** Logical CSS properties (**margin-inline-start**, not **margin-left**) everywhere. Physical properties require an RTL counterpart in the same commit.
16. **Embedded LTR runs inside RTL text are wrapped explicitly** (**dir="ltr"** spans) for phone numbers, emails, addresses, acronyms and codes. The Unicode bidi algorithm does not reliably keep them in reading order, and this produces subtly wrong phone numbers that nobody notices until a parent cannot call the school.
17. **Script-specific typography is engineered, not fallen back to.** Arabic renders in Amiri/Cairo, Yorùbá in a face with correct diacritic support. A Latin font's fallback for Arabic is a defect.
18. **Untranslated pages are derived with translated chrome and a labelled notice**, never omitted. A language switcher that points at a 404 is worse than an honest partial translation.
19. **Machine translation is never shipped as finished work.** AI-assisted, human-reviewed, and disclosed (Volume I §7.2).
20. **Locale-correct formatting** for dates (including Hijri where relevant), numbers, currency, names and address order. A Nigerian date rendered US-style is a small error that costs real trust.

CHAPTER 12 — SECURITY BIBLE I: PRINCIPLES & ZERO TRUST

12.1 The security principles

S1 — Assume breach. Design so that a compromised component does not compromise the system. **S2 — Least privilege, always, including for us.** A StromeX engineer’s default access to customer data is none. **S3 — Defence in depth.** No single control is load-bearing. **S4 — Secure by default.** The safe configuration is the one you get without asking. **S5 — Fail closed.** When an authorisation check cannot complete, deny. **S6 — Security is free** (Volume I §6.1). Patches, TLS, WAF, backups and audit logs are never a paid upgrade. **S7 — Children’s data receives the strictest handling in the group,** without exception and without a customer having to request it. **S8 — Disclose fast and completely.** Reputational damage comes from concealment, not from incidents.

12.2 Zero trust, concretely

PRINCIPLE	IMPLEMENTATION
No implicit network trust	Every service-to-service call is authenticated and authorised; being “inside” grants nothing
Verify explicitly	Identity, device posture and context evaluated per request, not per session
Least privilege	Short-lived, scoped credentials; no long-lived shared secrets; no shared accounts, ever
Micro-segmentation	Network policy between services; databases reachable only from their owning service
Continuous verification	Session re-evaluation on privilege change, anomalous location, or new device
Encrypted everywhere	TLS 1.3 in transit including internally; encryption at rest by default
Assume compromise	Anomaly detection on access patterns; blast-radius limits designed in

Table 11 — Zero trust, concretely

12.3 Staff access to customer data

- Default access: **none**.
- Access requires a ticket, a stated reason, a time limit, and it is **logged and visible to the customer** in their audit log.
- Production database access requires two-person approval and is session-recorded.
- Access to sensitive-class data (health, safeguarding, biometrics, children’s records) requires a named role, additional approval, and generates a customer-visible notification.
- Access reviews quarterly; departure revokes everything within one hour and this is tested.

CHAPTER 13 — SECURITY BIBLE II: IDENTITY & ACCESS

CONTROL	STANDARD
Password storage	Argon2id, per-user salt. Never anything else, never reversible
Password policy	Length over complexity; breached-password screening; no forced rotation without cause
MFA	Available on every tier including free; mandatory for admin, finance, credential-issuing and sensitive-data roles
Passkeys / WebAuthn	Supported and preferred; the recommended default as adoption allows
Sessions	Short-lived access tokens, rotating refresh tokens, server-side revocation, device list visible to the user
Token denylist	Immediate revocation capability — already implemented in the MVP
SSO	SAML 2.0 and OIDC for institutions
Authorisation	Role-based with attribute conditions; deny by default; evaluated server-side always
Privilege escalation	Requires approval; time-boxed; logged; auto-expiring
Service accounts	Scoped, rotated, never shared, never interactive
Account recovery	Verified out-of-band; recovery is the most-attacked path and is treated as such
Brute force	Progressive delays and lockout with a documented, non-punitive recovery route
Rate limiting	Per-identity and per-IP, at the edge and at the application

Table 12 — Security Bible II: Identity & Access

CHAPTER 14 — SECURITY BIBLE III: CRYPTOGRAPHY & DATA PROTECTION

CONCERN	STANDARD
In transit	TLS 1.3; HSTS with preload; modern cipher suites only; certificate transparency monitored
At rest	AES-256 volume and database encryption as the baseline
Field-level	Sensitive-class fields (health, safeguarding, biometric templates, bank details) encrypted at field level with separate key custody
Key management	HSM or managed KMS; documented rotation; two-person control for credential-signing keys (Chapter 5.3)
Signing	Per-institution key pairs; no group master key
Secrets	A secrets manager, never in source, never in environment files committed to a repository, never in logs. Scanned for in CI and in the repository history
Randomness	Cryptographically secure sources only
Hashing	SHA-256+ for integrity; Argon2id for passwords; never MD5 or SHA-1 for anything security-relevant
PII in logs	Redacted at the logging layer, not by convention
Backups	Encrypted, key-separated from the primary system
Biometrics	Templates only, never raw images ; encrypted; deleted on relationship end; never leave the institution's tenancy
Payment data	Never stored. Tokenised at the processor. PCI scope minimised deliberately
Quantum posture	Track NIST post-quantum standards; prioritise credential signatures for migration, since a certificate issued in 2027 may need to remain verifiable in 2067 and is therefore the group's longest-lived cryptographic commitment (Volume IX, Chapter 7)

Table 13 — Security Bible Iii: Cryptography & Data Protection

CHAPTER 15 — SECURITY BIBLE IV: SECURE DEVELOPMENT

PRACTICE	RULE
Threat modelling	Required for any feature touching auth, payments, credentials, or sensitive-class data
Code review	Every change, by someone who did not write it. No exceptions for seniority or urgency
SAST / DAST	In CI, blocking on high severity
Dependency scanning	Continuous; critical vulnerabilities patched within 48h, high within 7 days
Supply chain	Pinned dependencies; lockfiles committed; provenance verified; a new dependency requires a justification
Secrets scanning	Pre-commit and in CI, including history
Input validation	Server-side always; client-side is UX, never security
Output encoding	Context-appropriate; templating auto-escapes by default
SQL	Parameterised queries only. String-built SQL is a blocking review failure
SSRF	Allow-list outbound destinations; validate and re-validate redirects; block private address ranges. This is called out specifically because a critical SSRF vulnerability was found, reproduced and fixed during the group's own independent MVP audit — the lesson is retained here rather than quietly forgotten
File uploads	Type and size validated; content-sniffed; stored outside the web root; served from a separate origin; scanned
Deserialisation	Never deserialise untrusted input into executable structures
Error handling	No stack traces, internal paths, or version strings to users
Security headers	CSP, HSTS, X-Content-Type-Options, Referrer-Policy, Permissions-Policy — verified in tests
Penetration testing	Annual minimum; before any government or enterprise go-live; after any architectural change to auth or credentials
Bug bounty	Public programme from Phase I; safe harbour stated; researchers paid and credited

Table 14 — Security Bible Iv: Secure Development

CHAPTER 16 — SECURITY BIBLE V: RESILIENCE, BACKUP & CONTINUITY

16.1 Backup

PROPERTY	STANDARD
Frequency	Continuous WAL archiving + daily full
Retention	30 days standard; 1 year and 7 years available
Geography	At least two regions; at least one outside the primary jurisdiction
Encryption	Always, with separate key custody
Immutability	Write-once retention on at least one copy — ransomware defence
Restore testing	Monthly, automated, verified against a checksum and a functional smoke test. An untested backup is not a backup and may not be described as one

Table 15 — Backup

16.2 Recovery objectives

TIER	RPO	RTO
Standard	24h	8h
Professional	4h	4h
Premium	1h	2h
Elite	15 min	1h
Enterprise	5 min	30 min

Table 16 — Recovery objectives

16.3 Continuity

Documented and **rehearsed**: regional failure, database failure, provider failure, payment-processor failure, AI-provider failure, key-person unavailability, office inaccessibility, and extended power or connectivity loss in a market. A plan that has never been rehearsed is a document, not a capability. Rehearsal cadence: at least annually, with findings recorded and acted on.

CHAPTER 17 — SECURITY BIBLE VI: INCIDENT RESPONSE

17.1 Severity

SEV	DEFINITION	RESPONSE
S0	Customer data exposed; credential system compromised; funds at risk	Immediate, all hands, executive informed within 15 min
S1	Platform down; auth broken; data integrity at risk	Immediate, on-call escalated
S2	Major feature broken; performance severely degraded	Same business day
S3	Minor defect with a workaround	Next release

Table 17 — Severity

Cross-tenant data access is automatically S0, regardless of how many records were involved, including one.

17.2 The response sequence

Detect → declare (anyone may declare; nobody is penalised for a false alarm) → assign an incident commander → contain → preserve evidence → **notify the customer** → eradicate → recover → verify → **publish**.

17.3 Notification commitments

AUDIENCE	TIMING
Affected customers	Within 24 hours of confirmation , before we fully understand it, saying what we do and do not yet know
Regulators	Per statute (NDPA, GDPR: 72 hours)
Public	With the post-incident review
Data subjects	Where required by law, or where the risk to them is material regardless of legal requirement

Table 18 — Notification commitments

17.4 The post-incident review

Published, blameless, and specific. It states what happened, the timeline, the impact including the number of records, the root cause, what we fixed, and what we changed so it cannot recur. It names no individual. It does not use the passive voice to obscure agency (Volume I §11.2.4).

We publish these even when nobody would have found out. The willingness to do so is the asset; the incident is just an event.

CHAPTER 18 — QUALITY, TESTING & THE DEFINITION OF DONE

18.1 The test pyramid

Unit (fast, numerous) → integration (real database, real dependencies) → contract (API compatibility) → end-to-end (critical journeys only) → performance (against Chapter 10 budgets) → security (the cross-tenant suite, auth suite, injection suite) → accessibility (automated, plus manual on new patterns) → **restore** (Chapter 16.1).

Mandatory suites that may never be skipped in CI: cross-tenant isolation, authentication and authorisation, payment idempotency, credential signing and verification, and data export completeness.

18.2 The Definition of Done

A change is done when **all** of the following are true:

21. It works on a mid-range Android on 3 Mbps.
22. It works in RTL and in at least one non-English locale.
23. It meets WCAG 2.2 AA, verified.
24. It is within the Chapter 10 performance budgets.
25. It has tests at the appropriate levels, and they pass.
26. It has been reviewed by someone who did not write it.
27. Its failure modes are defined and its degraded mode is implemented.
28. It is observable — logs, metrics, and an alert if it matters.
29. It is documented where a user or an integrator would look.
30. It exports. Any new record type is in the export.
31. It has a rollback, and the rollback has been considered rather than assumed.
32. Its audit events are emitted.
33. A migration, if any, has a tested reverse path.

“Done” is not “merged”. It is not “works on my machine”. It is not “we’ll add tests later”.

CHAPTER 19 — RELEASE ENGINEERING

Trunk-based development with short-lived branches · every commit builds and tests · staging mirrors production in shape if not scale · progressive rollout (internal → 5% → 25% → 100%) with automatic rollback on error-rate or latency regression · feature flags for anything user-visible and risky · database migrations are backward-compatible and deployed separately from the code that requires them · **never deploy on a Friday afternoon or during an institution’s examination window, results period, or admissions deadline** — the customer’s calendar constrains our release calendar, and this is a hard rule that has cost us less than one incident would have.

Release cadence: continuous for fixes, weekly for features, quarterly for anything requiring customer preparation. Institutions receive 30 days’ notice of any change that alters a workflow they have trained staff on.

CHAPTER 20 — OBSERVABILITY

Structured logs with correlation IDs and redacted PII · RED metrics (rate, errors, duration) per service and USE metrics (utilisation, saturation, errors) per resource · distributed tracing across service boundaries · real user monitoring from the actual devices and networks our users have, not synthetic tests from a data centre · error tracking with grouping and ownership · a **public** status page with real uptime and full incident history · alerting that pages a human only for things a human must act on immediately, because an alert nobody acts on trains everyone to ignore alerts.

The customer-visible half: every institution sees its own audit log, its own uptime, its own SLA credit position (applied automatically — Volume III §17.2), and its own consumption against caps.

CHAPTER 21 — TECHNICAL DEBT & ARCHITECTURE DECISION RECORDS

21.1 ADRs

Every architecturally significant decision is recorded: context, options considered, decision, consequences, and what would make us revisit it. Stored in the repository, numbered, immutable once accepted, superseded rather than edited. A decision whose reasoning is undocumented will be re-litigated by someone who does not know it was already settled — that is the actual cost this prevents.

21.2 Debt

Debt is **registered, not tolerated silently**. Each item carries its cost (what it slows or risks), its interest (whether it worsens with time), and a decision: pay now, pay by a date, or accept permanently with a stated reason. **20% of engineering capacity is reserved for debt, tooling and reliability** and is not reallocated to features under delivery pressure. Under sustained pressure, scope is cut before this reserve is — because the reserve is what makes the next quarter possible.

Glossary

Terms are defined as this corpus uses them. Where a term also has a general industry meaning, the definition here governs internally and the cross-reference points to the chapter in which the term is established.

TERM	DEFINITION
ACV	Annual contract value. The recurring revenue a customer contributes in a year, excluding one-time implementation and build fees.
Band A / B / C	The three regional price bands defined in Volume I §6.4. All catalogue prices are quoted at Band C (list); Band A applies a 0.32× multiplier and Band B a 0.65×. The band follows the institution's operating jurisdiction, not its billing address.
CAC	Customer acquisition cost. Loaded sales and marketing cost divided by customers acquired in the period.
CAC payback	The number of months of gross profit from a customer required to recover that customer's acquisition cost.
Craft standard	The requirement in Volume VI §2.1 that creative work be indistinguishable from that of a good London or New York studio while remaining reproducible by a regional printer on the first attempt.
Credential	Any institution-issued, cryptographically signed record of an award, qualification, identity or transaction that a third party may need to verify. The subject of Volume IV, Chapter 5.
Degraded mode	The defined, designed and tested behaviour of a surface under failure — no network, slow network, provider outage. Required of every surface by Volume IV §10.2.
Ecosystem	A complete sector solution spanning all eight architectural layers, rather than a product sold into a sector. The unit of strategic expansion; see Volume I, Chapter 2.
Entrenched provision	A clause amendable only by board supermajority with published rationale and thirty days' notice. Enumerated in Volume IX §9.2.
Federation	Server-to-server interoperation across independently operated instances, allowing an institution to own its data while remaining reachable. The basis of the SpaceTalk government strategy; Volume X, Chapter 5.
Foundation tier	The permanently free tier. Not a trial and not a demo; governed by the Free Constitution in Volume I, Chapter 5.
Gate	A set of conditions that must be met before a roadmap phase may end. A phase ends when its gate is passed, not when its years elapse. Volume VIII, Chapter 2.
GRR	Gross revenue retention. Recurring revenue retained from existing customers excluding expansion; measures churn alone.
Guilloche	Fine mathematical line-work used in security printing, extremely difficult to reproduce by scan-and-print. The primary visual security element of a StromeX credential; Volume VI §6.2.
Human-in-the-loop	The requirement that a named human decides, and is recorded as deciding, wherever the stakes ladder classifies a decision as high. Volume IV §8.1.
LTV	Lifetime value. The discounted gross profit expected from a customer over the expected life of the relationship.

TERM	DEFINITION
No-fork rule	The prohibition on forking shared platform layers for a single customer without written CTO approval and a sunset date. Volume IV §3.2.
NRR	Net revenue retention. Recurring revenue from existing customers including expansion and contraction. Above 100% means the existing base grows without new customers.
Pricing Council	The standing body that owns Volume III, meets quarterly, and reviews every price against delivered margin. Volume I §8.4.
Reference Implementation	A delivered institutional deployment used as proof of method and as a permanent sales asset. Four are recorded in this edition.
Rule of 40	Revenue growth percentage plus operating margin percentage. A common composite measure of whether a software company balances growth against profitability.
Stakes ladder	The four-level classification — low, medium, high, prohibited — determining how much autonomy an AI system may exercise over a decision. Volume IV §8.1.
System of record	The authoritative store of an institution's people, money, documents, credentials and decisions. Identified in Volume II §9 as the group's only genuinely durable moat.
Value ledger	The running, honest account of what an institution has gained from StromeX, shown at every renewal including the periods where the numbers are unimpressive. Volume I §4.4.
Verification token	A prepaid unit purchased by an institution for verification-adjacent services carrying real marginal cost. Verification itself is always free to the verifier. Volume III §3.5.
Wave	A group of industry ecosystems opened in the same roadmap phase. Volume I §2.2.

Table 19 — Glossary of terms

References & Bibliography

Every external figure in this corpus is attributed at the point of use. This list consolidates those sources and records what each was used for. Sources were consulted in August 2026; where a source reports a market figure that moves, the figure is dated in the text.

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Internal sources

Claims in this corpus about what the group has built trace to these. Catalogue lines in Volume III are marked built, partial or specified-not-built on this basis.

StromeX MVP · [apps/api](#), [apps/web](#), [infra](#)

FastAPI backend with multi-provider AI routing, memory, spaced-repetition tutor and PDF export; Next.js frontend; Docker Compose infrastructure. Establishes the canonical stack in Volume IV §2.2.

Independent MVP Audit · [docs/04-STROMEX-INDEPENDENT-AUDIT.md](#)

Security, scalability, reliability and performance audit including a critical server-side request forgery vulnerability found, reproduced and fixed. Carried forward as a standing engineering rule in Volume IV §15.

Reference Implementation №1 · [Sultan Hanafi Royal Schools](#)

Four-language institutional build (English, Arabic RTL, Yorùbá, French) with approximately thirty-five page templates per locale; four verification surfaces; certificate generation with documented identifier architecture, security model and signing-key deployment; digital identity and card system; finance platform; eight role portals; personalisation centre; and approximately one hundred and twenty institutional system, governance and design documents.

Reference Implementation №2 · [WorldWide English College](#)

A seventy-three-table credential and commerce schema across fifteen migrations, covering enrolment integrity, competencies, evidence, awards, credential signing, issued documents, verifying institutions, payments, instalment plans, multi-currency routing, corporate seats and alumni chapters. The evidential basis for Volume IV §4.2.

Reference Implementation №3 · [Al-Madeenah College](#)

Bilingual static build system with offline service worker, verification surfaces, font subsetting pipeline and document generation to PDF and DOCX.

Reference Implementation №4 · [Institutional portal system](#)

Multi-role portal implementation covering student, staff, parent, applicant and administrative roles, admissions pipeline, attendance, timetable, assessment, finance and library.

Index

Generated by the build. Page references point to where each concept is established or substantively discussed; passing mentions are not indexed. Word index markers are also embedded throughout the corpus, so a live index can be built in Word if preferred.

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Colophon

This edition is composed for A4 at 210 × 297 mm. Display matter is set in a serif book face and running text in a companion serif, with a sans for tables, captions and navigation and a monospaced face for code and architecture figures — the four typographic roles defined in Volume I, Chapter 10. The StromeX brand faces, **Archivo** and **Fraunces**, ship with this edition in [docs/bible/publication/fonts/](#); the build selects them automatically wherever they are installed in the rendering environment, so the corpus can be reissued in full brand typography without editing a line of it.

The rules, tables and figures follow the design laws in Volume I, Chapter 12: typography carries the interface, space is a component, colour carries meaning, and structural devices encode something true about the content rather than decorating it. Tables use horizontal rules only; the vertical grid is carried by alignment, as it is in every well-set financial document.

This DOCX is the authoritative master. The press-quality PDF edition is generated from it and is content-identical. Both are produced from the markdown corpus of record under version control at [docs/bible/](#), by the build recorded at [docs/bible/publication/](#), so that the publication can be regenerated exactly whenever the corpus is amended.

THE STROME X EDITORIAL BIBLE · Edition II ·
VERSION 2.0

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